Year 9 Science

Plate Tectonics

Plate tectonics is the theory that the earth's crust is divided into several plates (about 30). It is the movement of these plates that caused Pangaea to break apart, and continues to cause the drifting of the continents. But what then causes the tectonic plates to move?

Watch the video carefully and answer the following questions. You also need to use your textbook (Chap 8.2, page 296 – 298) for some of these answers.

Earth's layers

1. Which two elements make up most of the inner core?

   _____

2. Which layer is the thickest?

   _____

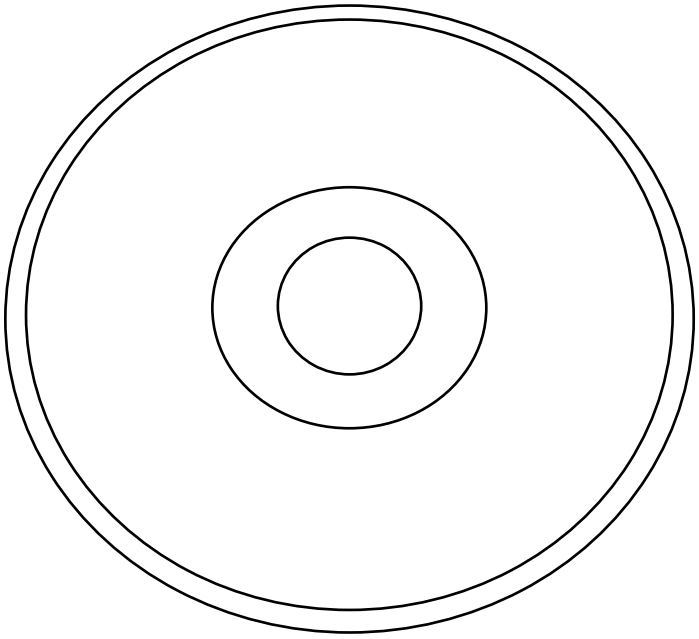
3. The upper mantle and crust make up the _____sphere.
4. Describe the appearance of the earth's outer shell.

   _____

   _____

Movement of tectonic plates

1. Using the diagram, sketch what happens in the mantle that causes the movement of the tectonic plates of the upper mantle and crust.



https://www.youtube.com/watch?v=0mWQs1_L3fA

2.

a. Describe a divergent boundary.

_____

b. Divergent boundaries are also called _____ plate boundaries because _____

_____

c. What are ocean ridges and how are they formed?

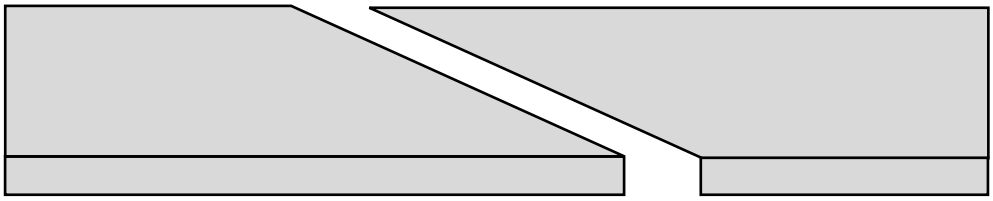
_____

_____

_____

d. Show or sketch **and label** on the diagram below:

- The direction of movement of divergent tectonic plates.
- Rising magma from the mantle
- Ocean ridges.



3.

a. Describe a convergent boundary.

_____

b. Convergent boundaries are also called destructive plate boundaries because

_____

_____

c. Subduction is when oceanic crust

_____

_____

_____

d. How are volcanoes formed?

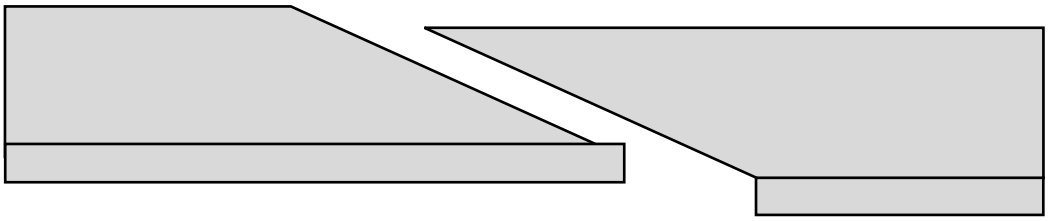
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e. Show or sketch **and label** on the diagram below:

- The direction of movement of convergent tectonic plates
- The oceanic plate subducting under the continental plate
- Solid crust becoming molten rock in the mantle
- The land formation often associated with convergent boundaries



4. (textbook)

a. Convergent boundaries can also occur between two continental plates. Which type of land formation often results from the collision of two continental plates? How does it happen?

_____

_____

_____

b. Name a well-known land formation that has resulted from this type of collision.

_____

5. (video and textbook)

a. Describe a transform boundary

_____

_____

b. Transform boundaries are also called conservative plate boundaries because

_____

_____

c. Think about the word ‘transform’. Why might the sliding segments of tectonic plates be called ‘transform boundaries’?

_____

_____

d. Do transform boundaries pose any dangers to humans and other creatures? Explain.

_____

_____

_____

e. Show or sketch **<u>and label</u>** on the diagram below:

- The direction of movement of transform plate boundaries
- Land formations that could be associated with transform boundaries

